Md. Minaruzzaman Shovon

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RESEARCH INTERESTS

Coastal shoreline dynamics & forecasting; machine learning & remote sensing for water systems; hydrology & urban drainage modelling; water network resilience & security.

EDUCATION

Chittagong University of Engineering and Technology (CUET) *2023 – Present*
B.Sc. in Water Resources Engineering - Chittagong, Bangladesh

Chittagong College *2019 – 2021*
Higher Secondary Certificate (HSC) - Chittagong, Bangladesh

Chattogram Cantonment Public College
Secondary School Certificate (SSC) - Chittagong, Bangladesh

RESEARCH EXPERIENCE

Undergraduate Thesis - Department of Water Resources Engineering, CUET *2026 – Present*
Shoreline dynamics along a tropical delta coast using statistical and machine learning models
Supervisor: Md. Ashraf Khan. Extracting annual shorelines (1995–2025) for Polder 56/57, Bhola Island (Lower Meghna Estuary) from Landsat and Sentinel-2 imagery using CoastSat and DSAS; observed maximum erosion of −144.5 m/yr and accretion of +55.2 m/yr (LRR). Forecasting shoreline change with statistical (Kalman filter, ARIMA) versus machine learning (Random Forest, XGBoost) models under held-out hindcast validation, with explainable-AI analysis of wave, water-level, cyclone and discharge drivers.

Research Project - Bangladesh Coastal Polders Geospatial Dataset *2026*
Curated and published an open-access vector dataset of Bangladesh's 139 coastal polders (Bangladesh Water Development Board source data) on Google Earth Engine, standardized to WGS84 with sample code and shapefile downloads for coastal and delta researchers.

PUBLICATIONS

Conference Papers

1. "Analysis and Prediction of Time Series Variations in Rainfall by Machine Learning Approach: A Case Study of Sylhet and Sreemangal." Chowdhury, M. O., Kabir, A., **Shovon, M. M.**, & Chowdhury, P. *International Conference on Civil Engineering Research & Innovations (ICCEI)*, December 2025.

2. "Seasonal Anomaly Detection in the Halda River Using a Multivariate Deep Learning Framework." **Shovon, M. M.**, Tamjid, H. A. H., & Tamim, M. *International Conference on Civil Engineering Research & Innovations (ICCEI)*, December 2025.

Preprints

3. "Analyzing Seasonal Variations in Air Quality with Google Earth Engine: A Case Study of Chattogram, Bangladesh." Shovon, M. M., & Tamim, M. *EarthArXiv* (preprint), 2026. DOI: 10.31223/X5S767.

4. "A Compact Open Encoder Model for Calibrated Typed Decisions on Unseen Tasks." Shovon, M. M. *Research Square* (preprint), 2026. DOI: 10.21203/rs.3.rs-11163029/v1. (Under review at *Discover AI*.)

5. "Downstream Constrained Dynamic Evacuation of Municipal Stormwater Detention Networks: A Real Time Feedback Control Approach." Shovon, M. M. *SSRN* (preprint), 2026. DOI: 10.2139/ssrn.7635154. (Under review at *Journal of Hydrology*.)

6. "Domain-Adaptive Fine-Tuning of a 1.7B LLM for Security Event Classification and SOC Triage." Shovon, M. M. *Research Square* (preprint), 2026. rs-9862964.

Manuscripts Under Review & Submitted

7. "Sample-Efficient Machine Learning Surrogates for Failure-Aware Resilience Screening in Pumped Urban Drainage Systems." Shovon, M. M. Under review at *Urban Water Journal*, 2026.

8. "Cyberattack Detection in Water Distribution Networks Under Sensor and Model Uncertainty." Shovon, M. M. Under review at *IJCIP*, 2026.

9. "Risk Informed Seismic Retrofit Prioritization for Urban Water Distribution Networks." Shovon, M. M. Under review at *Water Research*, 2026.

PROFESSIONAL EXPERIENCE

Authry AI *Sep 2025 – Present*
Chief Technology Officer - Dhaka, Bangladesh
Built a deepfake detection model and developed a full web application that detects deepfakes and fake news instantly.

Blendin (blendin247.com) *Aug 2025 – June 2026*
Web Developer - Chittagong, Bangladesh
Built production web applications and ERP systems using WordPress, Next.js, and Node.js.

Logifu Pty Ltd *Oct 2025 – Feb 2026*
Senior Software Developer
Delivered full-stack web development work as a senior developer within the engineering team.

SECURITY RESEARCH

13 CVE records credited on cve.org (WPScan and Patchstack CNAs), plus a changelog-credited disclosure in WP Store Locator.

1. CVE-2026-10749 - PHP Object Injection in Post Duplicator (CVSS 7.2, High).

2. CVE-2026-14322 - Unauthenticated Booking Auto-Approval in Timetics (CVSS 5.3, Medium).

3. CVE-2026-14556 - Unauthenticated HTML Injection in HappyForms via unsanitised referral value in notification emails (fixed in 1.26.15).

4. CVE-2026-14821 - Contributor+ Arbitrary Template Deletion in Quiz And Survey Master (CVSS 2.7, Low).

5. CVE-2026-14822 - Unauthenticated PayPal Order Status Manipulation in Event Tickets.

6. CVE-2026-15048 - Unauthenticated Sensitive Information Exposure in GeekyBot (CVSS 5.3, Medium).

7. CVE-2026-15049 - Arbitrary File Upload in Depicter via missing file-type validation in the import feature.

8. CVE-2026-15151 - Missing Authorization via rtb_reset_notifications in Five Star Restaurant Reservations (CVSS 3.8, Low).

9. CVE-2026-39756 - Unauthenticated IDOR in Wappointment (CVSS 6.5, Medium).

10. CVE-2026-57661 - Broken Access Control in WPComplete (CVSS 5.4, Medium).

11. CVE-2026-84021 - XSS in Bold Page Builder via improperly validated link URL output in an HTML attribute.

12. CVE-2026-85678 - Stored XSS in AI Builder via unsanitised custom JavaScript saved against a post.

13. CVE-2026-105998 - Unauthenticated Authorization Bypass in Booking Package via premium subscription reset (fixed in 1.7.31).

14. `WP Store Locator (≤ 2.3.0)` - Stored XSS (CVSS 8.1, High); patched in 2.3.1 and credited in the changelog.

MEDIA COVERAGE

The Daily Star, "Digital resistance: The websites that kept the revolution alive" (2024) - featured for co-developing **The Inquilab**, an archive of the July 2024 protests.

HONORS & AWARDS

1. 1st Place, AutoCAD Design Competition - Department of Water Resources Engineering, CUET, 2023

2. 2nd Runner-Up, IEEE CS Hackathon - IEEE Computer Society, BUET, 2024

3. Local Qualifier, NASA Space Apps Challenge, 2024

4. 11th Place, NextGen Hackathon - International Islamic University Chittagong (IIUC), 2025

5. Finalist (Top 21 of 150+ teams), BCS ICT Fest - Bangladesh Computer Society, 2025

6. 246th of 760, KnightCTF - KnightSquad Capture the Flag, 2025

7. 26th of 106 teams, SMP CTF - SMP Capture the Flag, 2024

CERTIFICATIONS

1. Google Cybersecurity Professional Certificate - Google, 2025

2. Cybersecurity Attack and Defense Fundamentals - EC-Council, 2025

3. Python Development - Microsoft, 2025

4. GIS & Its Application - Enhancing Digital Government & Economy Project, 2025

5. CISSP Training Course - InfoSec (Coursera), 2025

TECHNICAL SKILLS

GIS & Remote Sensing: Google Earth Engine, QGIS, ArcGIS, Google Earth Pro, DSAS

Hydrology & Engineering: HEC-RAS, HEC-HMS, EPA SWMM, EPANET, ETABS, AutoCAD

Programming & ML: Python (NumPy, Pandas, scikit-learn, PyTorch, TensorFlow), MATLAB, CNN/ANN, U-Net, LLM fine-tuning

Cybersecurity: Vulnerability research & disclosure (13 CVEs), WordPress security, web application security

Web Development: Next.js, React, TypeScript, Node.js, NestJS, Express.js, Expo (React Native), Tailwind CSS, REST APIs

Databases & Tools: PostgreSQL, MongoDB, Firebase, Redis, SQL; Git/GitHub, GitLab, Docker

Writing & Documentation: LaTeX, technical writing

LEADERSHIP & VOLUNTEERING

1. Water Resources & Research Organization Bangladesh - Joint Research Secretary, 2025; President, Dec 2025 - Present

2. Cyber Security Club, CUET - President, 2025–2026

3. Nogorful Bangladesh - ICT Director, 2022–Present

4. Andromeda Space & Robotics Research Organization Bangladesh - Web Developer, 2023–Present

5. Ongko - Founder and President, 2019–2020